

AI-Guided Exploration of Arithmetic Dynamical Systems:

Constraints, Failure Modes, and Search Strategies toward Hilbert–Pólya Structures

Liang Wang

School of Artificial Intelligence and Automation, Huazhong University of Science and Technology

Wuhan 430070, P.R. China

wangliang.f@gmail.com

September 13, 2026

Conference-oriented research record; generic, non-venue-specific format

Abstract

Recent advances in AI-assisted mathematical discovery, program search, and formally verified reasoning motivate a practical question: how can a long-horizon open problem be reorganized into a search process that is broad, falsifiable, and accountable to mathematical evidence? This paper presents a source-bound Phase-I case study around the search for arithmetic dynamical structures relevant to the Hilbert–Pólya programme. It does not claim a proof of the Riemann Hypothesis, a completed Hilbert–Pólya realization, or a spectral identification of Riemann zeros. Its central methodological contribution is a human-governed, AI-executed framework: the mathematician fixes the prime-symbolic research origin and scientific authority; mathematician and AI jointly maintain a versioned `AGENTS.md` contract specifying the admissible search space, route dispositions, a breadth-first strategy, and explicit stop conditions; and the AI returns a structured evidence handoff at a named gate or token/compute boundary. Within that contract, the task becomes candidate engineering. A candidate must retain one fixed object identity while connecting an endogenous arithmetic source, primitive closed orbits and their repetitions, a dynamical zeta or determinant, and, only later, an operator-theoretic realization. Phase I preserved useful local structures—including local trace/operator constructions, a same-object Fredholm chain, periodic-orbit classifications, finite symbolic results, and continuous-time ownership controls—but no documented top-level direction supplied an integrated $A_0+A_1+A_2$ chain in one natural candidate. The resulting recommendation is not to combine those partial successes retrospectively. It is to widen the next candidate-architecture search while applying early, source-bound tests for arithmetic origin, orbit/repetition ownership, and a same-object determinant. Detailed research summaries and provenance boundaries are retained in the appendices and the versioned repository.

Guided search-space constraint. The proposed Round-2 candidate space is lineage-constrained rather than a random collection of dynamical systems: each proposed object must document its prime-symbolic ancestor, its deformation/lift/geometric-realization relation, and the arithmetic mechanism that is preserved. A generic map or operator to which primes are attached only afterward is not admitted as a main candidate.

Keywords: arithmetic dynamics; human–AI collaboration; Hilbert–Pólya programme; periodic orbits; dynamical determinants; research methodology; reproducibility

Scope and nonclaims. This manuscript is a conference-oriented synthesis and research record. It reports a methodology, source-bound local results, controls, and open obligations. It does *not* establish the Riemann Hypothesis, a prime-power trace formula, a completed- Ξ determinant identity, a self-adjoint Hilbert–Pólya operator, or a route verdict transferred across distinct objects.

1 Introduction

The Riemann Hypothesis (RH) remains an unusually demanding target for human–AI mathematical exploration. As of 13 September 2026, the Clay Mathematics Institute continues to classify RH as an unsolved

Millennium Prize Problem [2]. The challenge is not simply to generate a numerical resemblance or an appealing operator. In the candidate class and evidence-obligation architecture adopted here, a credible Hilbert–Pólya (HP)-oriented candidate is required to connect arithmetic information, periodic dynamics, analytic structure, and an operator-theoretic trace identity without changing the mathematical object along the way.

This difficulty is timely in light of recent, but carefully bounded, advances in AI-assisted mathematics. Machine learning has helped human researchers formulate and investigate conjectures in selected areas of mathematics [4]; evaluator-coupled language-model search has generated new constructions in constrained combinatorial and algorithmic settings [6]; neuro-symbolic systems have solved 25 of 30 translated IMO-AG-30 Euclidean-geometry problems [8]; and reinforcement learning has been used to search for Lean-verified proofs in a specialized olympiad-level setting [5]. These examples show that AI can contribute to workflows with clear representations, evaluators, and verification boundaries. They do *not* constitute a proof of RH, a solution of any other active Millennium Prize Problem, or evidence that a language model can replace mathematical validation.

The question addressed here is therefore methodological:

How can the search for an arithmetic dynamical structure relevant to the HP programme be reformulated as a broad, falsifiable, and auditable human–AI candidate-search problem?

The answer developed from the Phase-I record is *candidate engineering*. Rather than optimize a loosely specified analogy to the Riemann zeros, the researcher freezes a candidate’s identity and tests an ordered set of obligations: endogenous arithmetic source, primitive-orbit and repetition ownership, a dynamical zeta or determinant belonging to that same object, and only then the stronger analytic and operator-theoretic requirements. AI can help enumerate, compare, document, and challenge candidates; human researchers remain responsible for definitions, proof acceptance, claim authorization, and external communication.

This paper makes five contributions.

1. It puts a human-governed, AI-executed research framework first: mathematicians choose the prime-symbolic origin and scientific constraints; a jointly maintained AGENTS.md makes the search protocol explicit; and AI returns control at stated route or budget boundaries.
2. It presents the updated programme roadmap as an *obligation architecture*, not a cumulative progress chart.
3. It compresses the Phase-I six-direction record into an evidence landscape that distinguishes useful local output from a missing bridge or a non-transfer boundary.
4. It formulates the *same-object* invariant and a candidate card that makes hidden ownership changes visible early.
5. It motivates a transition from family scanning to a wider portfolio of candidate architectures, screened first at A0–A2 rather than repeatedly polishing unrelated A0/A1-limited systems.

The detailed summary of each direction is placed in appendix A; the repository remains the canonical navigation layer for the underlying source materials [9]. The present manuscript is a research record and methodology paper, not an independent re-proof of those local results.

2 A human-governed framework for AI-assisted arithmetic-dynamical search

Figure 1 is the paper’s core methodological contribution: a human-governed framework for bounded AI search, organized in three layers. *Layer 1* gives the mathematician responsibility for the prime-symbolic research origin, the scientific question, permitted arithmetic data, and the admissible search envelope. *Layer 2* turns those choices into a jointly drafted, human-approved AGENTS.md contract. *Layer 3* requires bounded AI execution, an explicit evidence handoff, and mathematician review before another round can be authorized.

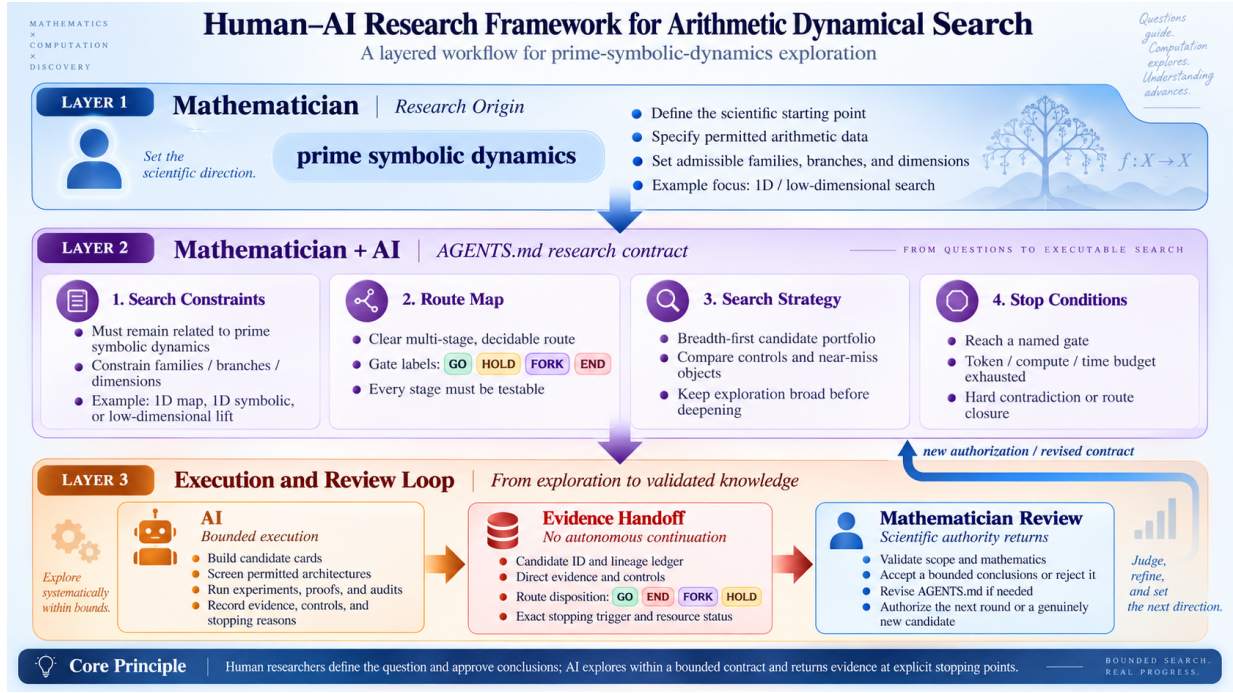


Figure 1: Three-layer human-AI research framework for arithmetic-dynamical search. *Layer 1:* the mathematician fixes the prime-symbolic origin and permissible search envelope. *Layer 2:* a jointly drafted, human-approved AGENTS.md fixes constraints, a testable route, a breadth-first portfolio with controls and near misses, and stopping conditions. *Layer 3:* bounded AI execution, evidence handoff, and mathematician review; continuation needs a new authorization or revised contract. GO, HOLD, FORK, and END are unordered workflow dispositions, not Route-A/Route-B passes. This is a prospective Round-2 framework, not a claim that Phase-I sessions used one uniform agent protocol.

This structure has a restrictive admission rule: no generic chaotic or quantum system may be admitted and decorated with primes later; nor may AI enlarge the admissible class, replace a missing preservation mechanism, or authorize a mathematical claim.

The Layer-2 contract has four irreducible parts; Layer 3 operationalizes them through bounded execution, evidence handoff, and mathematician review. First, the search-constraints card fixes the admissible class. For this programme, a main candidate must trace to a prime-symbolic ancestor and name a preservation, deformation, dimensional lift, or geometric-realization mechanism. It must also state the permitted system families, branches, dimensions, parameter sources, and arithmetic inputs before the search starts. The illustrative one-dimensional or low-dimensional focus in creffig:framework is a declared family restriction, not a license to attach prime language to any available dynamical system. Any later non-autonomous deformation or conservative lift must preserve, transform, or explicitly refute a named piece of that lineage.

Second, the route-map card contains a multi-stage route that can be decided at its stated scope. It specifies the evidence required for a gate, the controls that can falsify a proposed mechanism, and a non-ordered set of possible workflow dispositions: GO permits the next declared screen; HOLD returns an unresolved scientific decision to the mathematician; FORK creates a new object identity and candidate card; and END archives a bounded result. Every stage must be testable. These labels cannot be used as substitutes for A0–A4 or Route-B evidence.

Third, the search-strategy card makes breadth first the default. AI can compare several permitted architectures, including controls and near-miss objects, and keep source and ownership ledgers synchronized before deeper work begins. That breadth is disciplined: each architecture must pass the lineage admission test and cheap same-object checks. The strategy is not a claim that breadth is universally optimal; it is a practical response to the Phase-I finding that local successes from distinct objects do not add to one candidate.

Fourth, the stop-conditions card makes operational boundaries explicit. A search may stop at a named roadmap gate, a hard contradiction or declared route closure, a loss of lineage or same-object ownership, a scoped negative result, or exhaustion of its specified token, compute, or time budget. Budget exhaustion is an operational stopping condition, not a mathematical negative result. In Layer 3, AI may perform only contract-permitted candidate screening, experiments, proof attempts, and audits. At a stop it hands the mathematician a compact evidence packet containing the candidate ID, lineage ledger, direct evidence and controls, route disposition, exact stopping trigger, and resource status. The mathematician then validates scope and mathematics, accepts or rejects the bounded conclusion, and may issue a new authorization or a revised contract for a further round. There is no autonomous continuation. An AGENTS.md therefore provides reproducible research governance; it is not mathematical evidence and cannot validate a proof by itself.

The framework is prospective. The prior-work guide records a research trajectory and explicitly does not claim to have been a historical candidate selection manual. The framework uses that trajectory to govern future candidate engineering without retroactively assigning a uniform AI workflow to the six Phase-I directions [9].

3 From Hilbert–Pólya intuition to an obligation architecture

The framework in figure 1 governs how a candidate is admitted, screened, stopped, and returned to human judgment. The roadmap in figure 2 answers a separate question: which mathematical obligations an admitted candidate must earn. Neither diagram substitutes for the other.

The HP programme is often motivated by the expectation that a suitable spectral or trace structure could illuminate the nontrivial zeros of the zeta function. That motivation has inspired important connections among spectral asymptotics, trace formulae, and arithmetic [1, 3]. Dynamical zeta functions also have a well-established role in the study of expanding maps and Anosov flows [7]. None of these conceptual links licenses a candidate-specific conclusion by itself. The central practical question is which obligations must be tied to *one* object before an analogy can be treated as a serious candidate.

3.1 Route A and Route B

Figure 2 turns a diffuse analogy into a sequence of evidence obligations. Route A begins with A0, an endogenous arithmetic source. A1 requires primitive, reproducible orbit ownership together with a stated repetition convention. A2 asks for a stable dynamical zeta or Fredholm determinant whose operator, domain, normalization, and orbit expansion belong to the same candidate. A3 asks for the relevant global analytic or Weil-compatible structure, and A4 for a natural classical-to-quantum lift.

Route B is deliberately more demanding. It concerns a fixed operator and domain (B1), genuine self-adjointness (B2), the required spectral type (B3), an exact prime-power/Weil-compatible trace identity (B4), and a global completed- Ξ determinant or divisor equality (B5). It is not a second source of credit that can repair missing arithmetic or orbit ownership in a weak Route-A object.

The roadmap’s key word is *same*. A symbolic clock, physical closed orbit, determinant borrowed from a different transfer system, and separately constructed operator cannot be concatenated into one HP candidate. This is the *same-object* invariant:

Same-object invariant
Arithmetic source, clock/repetition, orbits, determinant, and operator must have one owner.

The Round-2 chain in the figure—prime arithmetic $\rightarrow$ symbolic structure $\rightarrow$ geometric carrier $\rightarrow$ closed-orbit / trace structure $\rightarrow$ quantum spectrum—is a proposed, constrained new-object design hypothesis. It is not a theorem-dependency graph, a declaration that the links already hold, or a request to attach results from separate Phase-I systems after the fact: each connection requires a new preservation ledger and *same-object* validation.

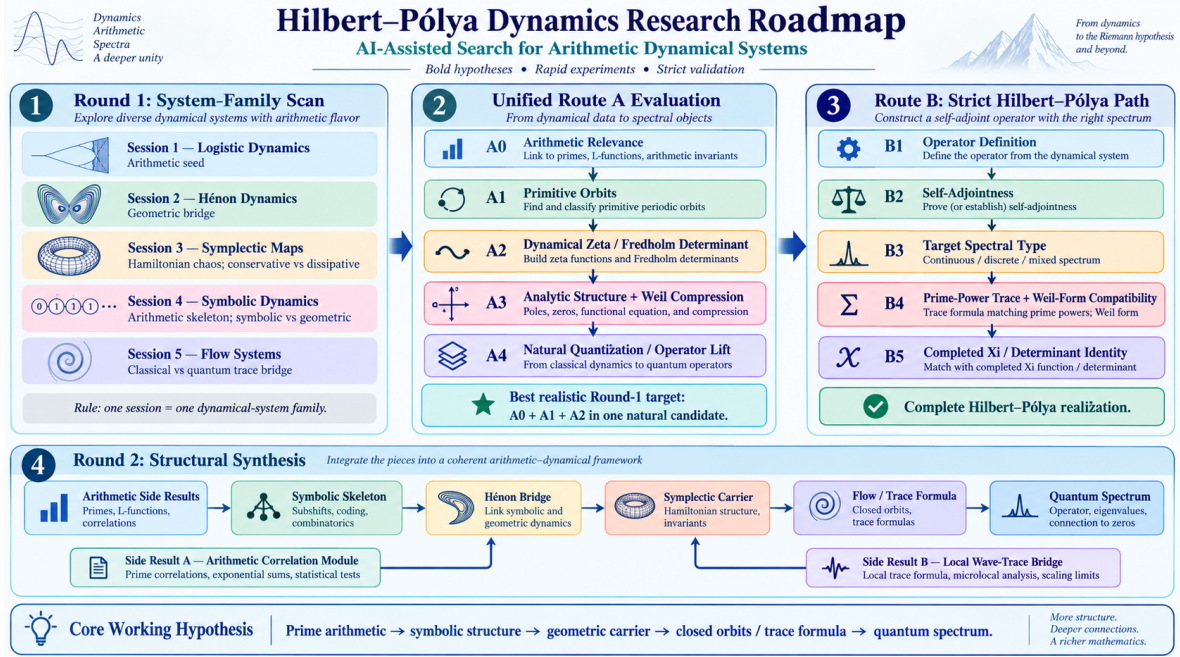


Figure 2: Author-proposed search and evidence-obligation roadmap. The updated roadmap separates a breadth-oriented Round-1 system-family scan, the Route-A candidate obligations, the stricter Route-B operator path, and a Round-2 structural-synthesis hypothesis. Arrows denote intended evidential continuity *within one candidate*; they do not report completed gates, transfer credit across systems, or establish an HP realization or RH. The Round-2 panel is a lineage-constrained new-object design hypothesis, not post-hoc assembly of Phase-I outputs; every link needs a preservation ledger and new *same-object* validation. The raster asset is a byte-preserved, release-local mirror of the user-supplied roadmap (SHA-256 f5e70c5120474702e4f0715bce90c7a88fd5936164594e8b37a2ee0993549d1d).

3.2 Why the workflow does not lower mathematical obligations

The practical value of AI in this setting is not an ungrounded “guess the answer” mode. It is most credible where the representation and rejection condition are exposed: candidate enumeration, source retrieval, object-ledger checking, adversarial control generation, proof-dependency review, and reproducible documentation. This perspective is close to the generator–evaluator separation demonstrated in program-search work [6], but the evaluator for a mathematical research candidate is partly formal and partly human. It includes source identity, mathematical scope, and whether a purported bridge preserves the object.

Human authority remains indispensable. A fluent synthesis can make nonidentical constructions look continuous; an automated retrieval system can surface a nearby theorem under a misleading normalization; a numerical fit can conceal target leakage. The human researcher must decide whether definitions are genuinely the same, whether a proof is valid, which claims have been earned, and whether a record may be released.

4 Phase-I record and source-bound evidence design

4.1 The Phase-I corpus and its status vocabulary

The Phase-I knowledge base separates one early free-exploration line, `zeta_mvp0`, from five roadmap-driven sessions: Logistic Dynamics, Hénon Dynamics, Symplectic Map, Symbolic Dynamics, and Flow Systems [9]. The five P24–P28 continuous-time forms belong to the Flow session; they are not an additional top-level direction. The detailed historical synthesis is bounded to the stated source-corpus snapshot 419ee36c1e310469209f7b83c096ec8aea448386 (13 September 2026 UTC). The new paper and the updated roadmap are later documentation artifacts and do not retroactively alter that historical record.

Four kinds of statements are kept distinct throughout this paper:

Kind	Meaning and non-equivalence
Source identity	Where a file, claim, source lock, or imported archive comes from; it is not independent mathematical verification.
Process / prove-nance	A build, receipt, frozen contract, or completed batch; it is not automatically a theorem, Route pass, or publication.
Local mathematics	A result for a named object and scope; it is not a global zeta/zero/HP conclusion.
Route assessment	A gate evaluation for a typed candidate; it cannot transfer to another object or session.

This separation is especially important for AI-assisted work. It prevents a model from turning a polished PDF, a large corpus, or a familiar keyword into an unauthorized conclusion. It also preserves the value of a local theorem or a scoped negative control when it does *not* close the full route.

4.2 Guided search space: prime-symbolic lineage

For the forward-looking Round-2 candidate-engineering protocol proposed here, the admissible main-candidate space is lineage-constrained. It turns the project's prior-work trajectory into an explicit admission rule:

Prime-symbolic candidate lineage
 prime/composite observables $\rightarrow$ symbolic coding and admissibility
 $\rightarrow$ justified sequential / non-autonomous deformation
 $\rightarrow$ Logistic prototype and one-dimensional obstructions
 $\rightarrow$ area-preserving Hénon dimensional/geometric lift
 $\rightarrow$ higher-dimensional conservative / symplectic carrier
 $\rightarrow$ later prime-side explicit-formula / operator-geometry benchmark.

The prior-work guide records a sequence of ideas rather than a completed theory or historical candidate-selection manual. This paper's admission rule is an author-proposed methodological use of that sequence. The arrows in the displayed lineage chain are therefore a prospective admission constraint and a search genealogy, not completed mathematical implications. For any proposed candidate, the record must identify its ancestral node or nodes, the transformation class being used, and the exact mechanism that is preserved, deformed, lifted, or refuted. The mechanism might be a specified prime-symbolic observable, admissibility rule, return structure, coding, clock, or a rigorously defined deformation thereof. If that relation is not yet proved, it must be marked OPEN; it cannot be silently replaced by a narrative similarity.

This adds a second protection to the *same-object* invariant. A generic area-preserving map, fashionable arithmetic space, trace formula, or operator does not become an arithmetic-dynamical candidate simply because primes can be attached to it later. Without a documented prime-symbolic lineage and preservation ledger, it is an *external control*, not an admitted main candidate. Likewise, a sequential/non-autonomous model must state its evolution law and show why it is more than a time-dependent fitting schedule. Prime-side explicit-formula and finite operator-geometry results are a benchmark for keeping prime powers visible, not permission to insert prime tables, von Mangoldt weights, or zero data by hand [9].

4.3 Operational audit trail and historical boundary

The workflow has two layers. The original research lines were created under their own mathematical and project protocols; this paper does not rewrite history by claiming that they all used one uniform AI procedure. For the prospective framework in figure 1, the following table makes the division of labor operational. AI can prepare and check a bounded search, while the mathematician controls its scientific scope, mathematical validity, and authorization to continue or release.

The record is intended to be inspectable rather than merely persuasive. The repository preserves the navigation pages, original project materials, conclusion boundaries, and accompanying research-record paper [9]. A clean build or an internally complete batch is therefore reported only as a process fact. It does not substitute for a proof check, external peer review, or a Route verdict.

Table 1: Operational division of labor under the prospective framework in figure 1. The AGENTS.md contract constrains AI action but does not replace mathematical validation.

Stage	AI-assistible work	Human responsibility
Origin and constraints	map prior work; surface permissible branches; draft candidate-card and protocol options	choose the prime-symbolic origin; define the scientific question, permitted data, families, branches, and dimensions
AGENTS.md contract	formalize constraints; check decision labels and stopping clauses for consistency	approve and version-freeze the operative contract, its gate meanings, and its authorization boundary
Bounded search	enumerate eligible architectures breadth first; map sources, owners, and controls; prepare evidence ledgers	validate definitions, proofs, computations, preservation claims, and any GO decision
Stop, handoff, and release	record exact triggers; prepare the decision packet, link/citation checks, and reproducible artifacts	accept or revise a bounded conclusion; authorize a fork, renewed work, claims, authorship, and release

5 Phase-I evidence landscape

The six lines produced different kinds of useful output. They cannot be ranked as six positions along one common route, because their objects, evaluation schemas, and scope boundaries differ. Figure 3 therefore uses colors to distinguish evidence roles, not to imply a success score: blue marks a preserved local capability, purple a local analytic or operator-related structure, gray a route-neutral or non-comparable state, and red a named missing bridge or stop boundary.

The portfolio supports two conclusions. First, Phase I did *not* identify an already formed candidate that can simply be extended to a proof. More precisely, within the source-bound record, no top-level direction is documented as carrying an integrated A0+A1+A2 chain in one natural candidate. This is a statement about the record and its named objects, not an impossibility theorem about every system in the corresponding family.

Second, Phase I did not amount to empty failure. It preserved reusable structures: a same-object analytic determinant chain in the Logistic line, periodic classification and geometric interfaces in the Hénon and Symplectic lines, exact finite-system controls in the Symbolic line, and continuous-time ownership tests in the Flow line. The methodological result is the identification of bridges that cannot be filled by prose alone: arithmetic source, clock/repetition convention, analytic owner, and operator owner must be frozen together.

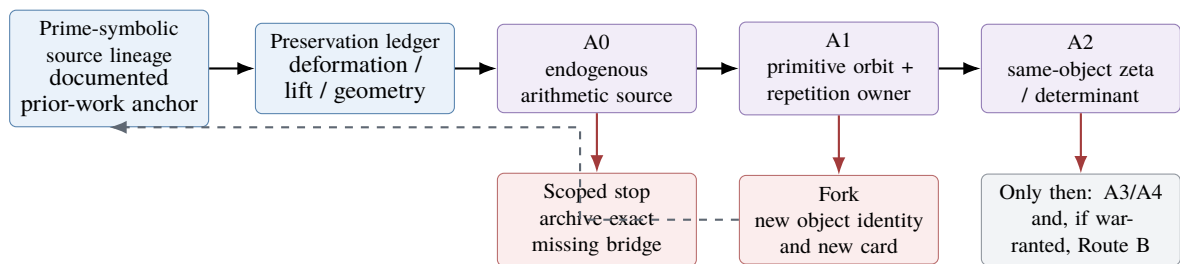
6 From family scan to candidate engineering

The principal Phase-I recommendation is not that breadth is universally superior to depth. The record is neither an optimization benchmark nor a controlled comparison of research strategies. Rather, repeated early boundaries, combined with the non-additivity of evidence across objects, motivate a change in emphasis: explore a broader portfolio of *candidate architectures*, but subject every architecture to low-cost, early, same-object tests before investing deeply in an A0/A1-limited system.

Here, broader does not mean a random sweep over generic dynamical forms. It means a portfolio of lineage-compatible descendants of the prime-symbolic starting point, each with a documented deformation, dimensional lift, geometric realization, or stated preservation failure. A system that lacks that entry condition remains useful as an external positive or negative control, but it does not enter the main candidate portfolio.

Direction	Preserved local capability	Next discriminating bridge / boundary
zeta_mvp0	local self-adjoint operator and relative-trace structures	endogenous prime-time / von Mangoldt bridge remains open
Logistic	same-object return–roof–transfer–Fredholm chain	arithmetic origin, Riemann-target closure, and quantization absent; root hunting stopped
Hénon	restricted periodic classifications and geometric diagnostics	A0/A1 weak or scoped; no documented A2–A4 closure
Symplectic	five local interfaces and diagnostic tools	Route applicability is NOT_APPLICABLE; freeze an actual candidate before gate claims
Symbolic	exact finite-system results and audit-friendly evidence pattern	no natural geometric carrier, global analytic owner, or operator closure
Flow	continuous-time clock, roof, trace, and ownership stress tests	for P24–P28: positive arithmetic A2 = 0/5; Route-B invocation = 0/5
Non-additive boundary: local results from distinct objects cannot be assembled retrospectively into one Hilbert–Pólya candidate.		

Figure 3: Phase-I evidence landscape: preserved capabilities, named gaps, and non-transfer boundaries. This is not a Route dashboard or a measurement of distance to RH. Each row reports a source-record category and the next discriminating issue for that line. In particular, NOT_APPLICABLE, internally complete, and scoped negative states are not comparable “scores.”



*AI can assist candidate enumeration, source mapping, contradiction checks, and control design;
human researchers own definition, mathematical validation, claim authorization, and release.*

Figure 4: From lineage-constrained family scanning to candidate engineering: proposed human–AI protocol. Every candidate enters through a documented prime-symbolic source lineage and must state its preservation, deformation, lift, or geometric-realization mechanism before A0 is discussed. The flowchart is a methodological recommendation from the Phase-I record, not an experimental proof that breadth dominates depth. A failed bridge is archived at its exact scope; a genuinely different construction is forked rather than inheriting credit.

Table 2: Candidate-card template for a lineage-constrained, *same-object* search. The card records early stop conditions rather than accumulated Route credit.

Field	Question and early stop condition
Object identity	What exact phase space, map or flow, parameter regime, measure, and normalization are under study? Stop if the object changes between sections.
Lineage and preservation	Which prime-symbolic prior-work object is the ancestor? Is this a non-autonomous deformation, conservative dimensional lift, or geometric realization; and which observable, coding, admissibility rule, or return structure survives? Stop before A0 if this ledger is absent; treat the system as an external control.
Arithmetic source	What internal mechanism produces the relevant arithmetic data? Stop if prime/zero data are inserted as an external selector.
Clock and repetition	What fixes a primitive orbit, period/roof/action, and the repetition law? Stop if a symbolic clock substitutes for a distinct physical clock without proof.
Analytic owner	Which transfer operator, zeta/determinant, domain, and normalization belong to this object? Stop if an analytic object is borrowed from a calibrator.
Controls	What nearby positive and negative controls distinguish a mechanism from a fitted coincidence? Stop if no falsifying comparison is available.
Operator owner	Which Hilbert space, operator, domain, and symmetry would support a later quantum claim? Defer, rather than assert, if they are absent.
Decision	Which gate is directly evidenced, open, not-testable, inapplicable, stopped, or forked? Never infer a label from adjacent work.

6.1 A candidate card before target matching

Every prospective candidate should receive a small, versioned candidate card before a Route label is discussed. The card in table 2 asks the following questions.

The card makes a crucial research economy possible. It places cheap, disqualifying ownership tests before expensive operator construction, long numerical searches, or a polished narrative. A Logistic-style analytic chain is then valuable as a control rather than as an invitation to resume closed root hunting; Hénon and Symplectic structures become possible geometric ingredients to test inside a new object, not inherited Route credit; and Symbolic coding is admissible only when it arises from the same candidate’s geometry and clock.

The same card also prevents a common historical reversal: select a generic high-dimensional or conservative system first, then decorate it with primes. For a non-autonomous proposal, the deformation law itself must be specified; for a dimensional or geometric lift, the card must name a projection, semiconjugacy, invariant, return relation, or explicitly open preservation claim. The words high-dimensional, area-preserving, or symplectic are not by themselves a prime-symbolic mechanism.

6.2 Why the next search should be architecturally broader

The roadmap’s Round-2 structural-synthesis direction gives a concrete shape to the proposal: seek a single candidate in which arithmetic structure, symbolic coding, a geometric carrier, closed-orbit/trace data, and any later quantum realization are derived from one construction. This does *not* mean splicing together a symbolic model, a symplectic map, a physical flow, and an operator with unrelated owners. It means designing a new object so that the relations can be checked rather than narrated.

There is an important distinction between wider search and lower standards. Candidate diversity is useful because Phase I exposed several orthogonal failure modes. The screening standard becomes stricter, not looser: every new architecture must first survive A0, then A1, then A2 with the same identity. An early scoped negative result is a successful search output when it rules out a named construction under named conventions; it prevents the next round from repeating the same ownership transfer in new vocabulary.

7 Discussion, limitations, and conclusion

The distinctive contribution of this work is methodological rather than a claim of mathematical closure. Its first contribution is the framework in figure 1: the mathematician fixes a prime-symbolic research origin and scientific authority; a jointly drafted, human-approved AGENTS.md contract makes the permissible search, decision labels, breadth-first strategy, and stopping conditions explicit; and AI returns a source-bound handoff rather than extending the project autonomously. The obligation roadmap then treats the HP programme as non-compensatory. A collection of impressive local features does not add up to a candidate. Negative controls, stops, and inapplicability become informative because they identify exactly where a bridge is absent. The versioned repository preserves provenance, source identity, and the distinction between historical research materials and this later synthesis.

Several limitations follow. The paper is not an independent proof audit of the component results; its detailed claims remain constrained by their named source records. It is not a benchmark comparing AI systems or research strategies. It does not establish that a broader portfolio is optimal, only that the Phase-I record supports a broader, gate-aware next search more strongly than repeated unbounded deepening of previously limited objects. Finally, it is not a proof of RH, a complete HP realization, an exact prime-power trace formula, a completed- Ξ determinant identity, or a spectral identification of Riemann zeros.

Within those limits, Phase I supplies a useful research result: it turns failure modes into design constraints. The next step is not to pick a winner from six incomparable lines. It is to construct and screen new *same-object* candidate architectures that begin from a documented prime-symbolic lineage, then state an explicit preservation/deformation/lift mechanism, arithmetic origin, clock/repetition ownership, determinant ownership, controls, and a route decision. If a future candidate survives those constraints, its positive claims will be easier to audit. If it fails, the recorded failure will still improve the next search.

Research materials and reproducibility

The versioned research record, protocols, source materials, roadmap asset, and buildable paper package are available at <https://github.com/maris205/hilbert-polya-structure> [9]. The historical detailed synthesis is tied to the source-corpus snapshot named in section 4; before an external submission, the author should create an immutable Git tag or release and cite that tag/commit in the venue-specific version. The repository URL alone should not be treated as a permanent archival identifier.

Ethics, authorship, funding, conflicts, and AI-use disclosure

Ethics. This work did not involve human participants, personal data, or animal research. **Author contributions (CRediT).** Liang Wang: conceptualization, methodology, investigation, visualization, writing—original draft, and writing—review and editing. **Funding.** No funding statement was supplied for this research-record version. **Competing interests.** A competing-interests declaration must be completed by the author before an external submission. **AI-use disclosure.** AI assistance was used for framework and protocol drafting, corpus navigation, source mapping, consistency-oriented drafting, visualization planning, and review-checklist preparation. The human author retains sole responsibility for mathematical interpretation, validation, claim approval, authorship, and release.

A Detailed Phase-I research record

This appendix keeps the direction-level detail that is deliberately compressed in the conference-oriented main text. Each paragraph is a source-bound summary of a named record, not a new theorem or a retroactive common Route evaluation.

A.1 Early free exploration: zeta_mvp0

The early zeta_mvp0 programme preserves local counting structures, a fixed self-adjoint compact-resolvent operator convention, and a certified local relative-trace construction. These are meaningful local operator and trace results. Its global claim ledger nevertheless leaves the endogenous prime-time mechanism open: no licensed internal rule supplies the relevant $r \log p$ -type timing with von Mangoldt weights. The corresponding zero-spectral relation and RH remain unauthorized. Thus an internal local label must not be translated into P1 Route-A credit merely because the vocabulary resembles a later route. The line also includes an imported RH archive whose source-preserving status is itself instructive: nearby material does not become native evidence merely by being stored in the same directory [9].

A.2 Session 1: Logistic Dynamics

The Logistic line supplies the clearest preserved example of a *same-object* analytic chain. At a frozen exact algebraic parameter, its record connects first-return structure, an intrinsic polar roof, complex inverse branches, boundary/partition bookkeeping, a nuclear transfer operator, a Fredholm determinant, and growth/order information. This is valuable precisely because the objects and normalizations are retained together. The same record also draws its boundary: the chain does not provide the Riemann-target arithmetic source, target-level global closure, or a natural quantization. Root hunting and Riemann-zero comparison are therefore stopped for that object. The methodological use of the line is as a seed/control for object ownership, not as an unfinished RH proof [9].

A.3 Session 2: Hénon Dynamics

The Hénon record studies restricted families through periodic classification and related low-/higher-dimensional geometric diagnostics. It preserves scoped rational/integer, affine-model, and period-spectrum results. Its Route-A interpretations remain object-specific: arithmetic relevance is weak; primitive-orbit information may be weak or fail for the particular object; and later analytic or lift obligations are absent or not testable in the named scope. Later frozen/admitted contracts were retained without being misrepresented as completed papers. This makes Hénon a potential Round-2 geometric bridge or Poincaré-section ancestor for a new construction, not an inheritance mechanism for a different candidate's analytic or arithmetic credit [9].

A.4 Session 3: Symplectic Map

The Symplectic session produced five scoped interfaces and diagnostic tools around conservative structures, periodic schemes, and qPI-related contexts. Its formal disposition is NOT_APPLICABLE for the route schema used elsewhere. That status is not a scientific failure, and it is not a pass. It means that no fixed candidate has been supplied to own simultaneously an endogenous prime-to-orbit mechanism, a logarithmic clock/repetition law, a target determinant, and a natural operator lift. The session is consequently valuable as a carrier/diagnostic resource, but it cannot be displayed as partial gate closure on a shared HP scorecard [9].

A.5 Session 4: Symbolic Dynamics

The Symbolic session provides exact, finite, autonomous deterministic-system results with a particularly audit-friendly evidence pattern. Its exact-five internal-complete and external-hold labels are process states, not a claim of global analytic continuation or an operator realization. The important structural boundary is that a finite symbolic system, however clean its algebra, does not automatically own a geometric carrier, physical clock, primitive closed-orbit convention, global determinant, or quantum operator. For Round 2, symbolic structure is therefore most promising when it is derived from a candidate's geometry rather than attached to a distinct object as an independent unit-roof model [9].

A.6 Session 5: Flow Systems

The Flow session turns the ownership issue into an explicit continuous-time stress test. Across its P24–P28 forms, it records local trace results, negative controls, and obstructions involving roofs, inverse limits, factorization, and operator ownership. Its bounded final ledger reports positive arithmetic $A2 = 0/5$ and Route-B invocation = $0/5$ for these five forms. The scope is crucial: this is neither a verdict on all flows nor a seventh top-level Phase-I direction. Its reusable conclusion is the strongest form of the *same-object* rule: one candidate must own clock, primitive orbit, repetition law, determinant, trace regime, and operator, rather than combining a symbolic clock, physical orbit, borrowed determinant, and separate operator [9].

B Candidate-card template and non-transfer checks

The compact candidate card in section 6 is the operational template. For the archival research record, its non-transfer checks are:

- C1. Identity.** Freeze phase space, map/flow, parameters, measure, and normalization. A changed object receives a new card.
- C2. Lineage.** Name the prime-symbolic ancestor and the precise deformation, dimensional lift, geometric realization, or preservation failure. Without this ledger, classify the system as an external control before A0.
- C3. Arithmetic and clock.** State permitted data, the endogenous arithmetic mechanism, primitive convention, time/roof/action, and repetition law. Do not replace a physical clock with a distinct symbolic clock.
- C4. Analytic and operator owners.** Name the transfer operator, zeta/determinant, domain, normalization, and any later Hilbert-space/operator pair. Do not borrow these from a proxy.
- C5. Controls and decision.** Record positive and negative controls, direct evidence, and an explicit OPEN, STOP, FORK, or EXTERNAL CONTROL decision. No neighboring Session supplies credit.

C Evidence register and asset provenance

Repository and source boundary. The public research record is <https://github.com/maris205/hilbert-polya-structure>. The detailed Phase-I synthesis is source-bound to commit 419ee36c1e310469209f7b83c096ec8aea448386; the new conference-oriented package is a later, separately versioned presentation layer. The repository’s P1 Wiki provides the six-direction index, claim vocabulary, cross-stream non-transfer rules, source cards, and links to the original project materials [9].

Prior-work lineage guide. The prime-symbolic search origin is documented in `flow_systems/docs/prior_work/README.md` and mirrored for the Round-2 workspace at `arithmetic_symplectic_flow/docs/prior_work/README.md`. These guides are used here as internal provenance for the constrained search genealogy, not as proof that any Logistic, Hénon, or symplectic candidate already realizes prime dynamics.

Research-framework asset. Figure 1 uses `assets/research_framework.png`, a byte-identical release-local mirror of the author-supplied P1 Wiki source `p1_wiki/research_framework.png`. Its SHA-256 is

```
da9e89f253991fb02db9ae11c2b13a51
2a4a404f720bf9abe03ff3853887f4a8
```

The diagram represents an author-proposed prospective governance framework. It is not evidence for a mathematical claim, a Route verdict, or a statement that Phase-I projects historically shared one workflow contract.

Roadmap asset. Figure 2 uses `assets/rh_roadmap.png`, a byte-identical mirror of the tracked Round-2 roadmap reference. Its SHA-256 is

f5e70c5120474702e4f0715bce90c7a
88fd5936164594e8b37a2ee0993549d1d

The asset is identified as author-proposed and is not substituted silently for the older Phase-I roadmap image used by the historical internal working paper.

External sources. The external bibliography supports only the background and methodology claims made in the main text: AI-assisted conjecturing, evaluator-coupled program search, bounded neuro-symbolic geometry, Lean-grounded proof search, Hilbert–Pólya/trace-formula context, dynamical zeta context, and the current official RH status. Detailed local mathematical statements are sourced to the repository record rather than inferred from the external references.

References

- [1] M. V. Berry and J. P. Keating. The riemann zeros and eigenvalue asymptotics. *SIAM Review*, 41(2):236–266, 1999. doi: 10.1137/S0036144598347497. URL <https://doi.org/10.1137/S0036144598347497>.
- [2] Clay Mathematics Institute. The millennium prize problems. <https://www.claymath.org/millennium-problems/>, 2026. Accessed September 13, 2026.
- [3] Alain Connes. Trace formula in noncommutative geometry and the zeros of the riemann zeta function. *Selecta Mathematica*, 5:29–106, 1999. doi: 10.1007/s000290050042. URL <https://doi.org/10.1007/s000290050042>.
- [4] Alex Davies, Petar Velickovic, Lars Buesing, et al. Advancing mathematics by guiding human intuition with AI. *Nature*, 600:70–74, 2021. doi: 10.1038/s41586-021-04086-x. URL <https://doi.org/10.1038/s41586-021-04086-x>.
- [5] Thomas Hubert, Rishi Mehta, Laurent Sartran, et al. Olympiad-level formal mathematical reasoning with reinforcement learning. *Nature*, 651:607–613, 2026. doi: 10.1038/s41586-025-09833-y. URL <https://doi.org/10.1038/s41586-025-09833-y>.
- [6] Bernardino Romera-Paredes, Mehran Barekatin, Alexander Novikov, et al. Mathematical discoveries from program search with large language models. *Nature*, 625:468–475, 2024. doi: 10.1038/s41586-023-06924-6. URL <https://doi.org/10.1038/s41586-023-06924-6>.
- [7] David Ruelle. Zeta-functions for expanding maps and Anosov flows. *Inventiones Mathematicae*, 34:231–242, 1976. doi: 10.1007/BF01403069. URL <https://doi.org/10.1007/BF01403069>.
- [8] Trieu H. Trinh, Yuhuai Wu, Quoc V. Le, He He, and Thang Luong. Solving olympiad geometry without human demonstrations. *Nature*, 625:476–482, 2024. doi: 10.1038/s41586-023-06747-5. URL <https://doi.org/10.1038/s41586-023-06747-5>.
- [9] Liang Wang. Hilbert–pólya structure: Phase-I research record and source materials. GitHub repository, <https://github.com/maris205/hilbert-polya-structure>, 2026. Accessed September 13, 2026.